

**QUANTITATIVE ECONOMICS - FALL 2023**  
**Answer any FIVE questions.**

1. Suppose you have the nonlinear regression

$$y_i = e^{\beta x_i} + u_i, \quad E[u_i | x_i] = 0$$

where  $u_i$  are zero mean IID error term and IID data  $\{(x_i, y_i)\}$  for  $i = 1, \dots, n$ . Describe in detail how you could estimate  $\beta$ ?

2. Let  $\hat{\beta}_1$  and  $\hat{\beta}_2$  denote the estimated regression coefficients for the linear model  $y = X_1\beta_1 + X_2\beta_2 + e$ , where  $X_1$  is an  $n \times k_1$  matrix,  $X_2$  is an  $n \times k_2$  matrix, and  $\beta_1$  and  $\beta_2$  are of dimensions  $k_1$  and  $k_2$ , respectively.

- (a) Show that  $\tilde{\beta}_1 = \hat{\beta}_1 + (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2$ , where  $\tilde{\beta}_1$  is a vector of coefficients from the regression of  $y$  solely on  $X_1$ .
- (b) Show that  $\hat{\beta}_1$  is an unbiased estimator of  $\beta_1$ .
- (c) Under what conditions  $\tilde{\beta}_1$  is an unbiased estimator for  $\beta_1$ .

**Hint:** Note that  $\hat{\beta}_1 = \tilde{\beta}_1 - (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2 = (X_1'X_1)^{-1}X_1'(y - X_2\hat{\beta}_2)$ .

3. Consider the following data generating process

$$Y_t = \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2}$$

where  $\varepsilon_t$  is a white noise process. Assuming  $0 < \theta_1 < 1$  and  $0 < \theta_2 < 1$  answer the following:

- (a) Derive the expected value of this process.
  - (b) Derive the variance of this process.
  - (c) Derive the covariance of this process for  $j = 1, 2, 3$ .
  - (d) Derive the autocorrelation function of this process for  $j = 1, 2, 3$ .
4. Consider two processes  $y_t$  and  $x_t$  where we have  $t = 1, 2, \dots, T$  observations on each. Suppose that  $y_t$  represents a 10 year interest rate and  $x_t$  represents a three month interest rate. Suppose we are interested in eventually forecasting  $y_t$  with the help of both past values of  $y$  as well as past values of  $x$ . With this information, answer the following questions:
- (a) Write down an ARMA(2,2) which also includes a lagged value for  $x$ .
  - (b) How do you know how many lags to include for  $y$ ? How do you know how many lags to include for  $\varepsilon$  (the error term)?
  - (c) How do you know how many lags to include for  $x$ ?
  - (d) Write down the  $h$ -step ahead value for  $y$  ( $y_{t+h}$ ) that you put down in part (a).
  - (e) For  $h = 1$ , construct the forecast value of  $y$  ( $\hat{y}_{t+h|t}$ ).
5. In 2019, City X and City Y were two neighboring cities with similar economies. In 2020, City X implemented a significant increase in the minimum wage, while City Y did not make any changes to its minimum wage policy. Researchers collected employment data from a random sample of restaurants in both cities in the years before and after the minimum wage change to examine its effects on restaurant employment.
- (a) Using a difference-in-differences (DiD) identification strategy, formulate a linear probability model (LPM) to investigate whether the minimum wage increase in City X had an impact on the probability of restaurant employees being employed in 2021. Which coefficient in your model quantifies the effect of the minimum wage change?

- (b) Interpretation of the Coefficient of Interest: How would you interpret the coefficient associated with the minimum wage change in your LPM?
- (c) Logit Model: Explain how the interpretation of the coefficient related to the minimum wage change would differ if you used a Logit model instead of an LPM.
- (d) Parallel Trend Assumption: Define the Parallel Trend Assumption in the context of this study. What strategies or methods could you use to assess whether this assumption is satisfied using the data from the restaurants in City X and City Y?

6. You are given the regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

You want to test the null hypothesis

$$\begin{aligned} H_0 &: \beta_1 \beta_3 = \beta_2^2 \\ H_A &: \beta_1 \beta_3 \neq \beta_2^2 \end{aligned}$$

Show how you would construct the test statistic.

- 7. You are researching the relationship between having health insurance coverage (explanatory variable) and the frequency of medical services utilization (dependent variable) among individuals. However, you suspect that unobserved factors affecting individuals' health status and behaviors could lead to endogeneity. To address this issue, you decide to use an instrumental variable.
  - (a) Propose an instrumental variable and a 2SLS regression model that enables you to estimate the causal effect of having health insurance on medical services utilization. Write down both the first stage and reduced form equations and incorporate control variables in your model.
  - (b) Show that reduced-form equation, can be derived by substituting first stage equation, into the causal relation of interest.
  - (c) Two Requirements of a Good IV: Explain two essential requirements that an instrumental variable must meet in your model. Explain how your suggested instrument for this scenario satisfies these two requirements.
- 8. You are studying the impact of a merit-based scholarship on the graduation rate of college students. The scholarship is awarded to students who achieve a GPA of 3.5 or higher during their first year of college. You suspect that receiving the scholarship might influence students' academic performance.
  - (a) Propose a sharp regression discontinuity model to estimate the causal effect of receiving the scholarship on the graduation rate. Please include any relevant control variables in your regression model and clearly define the treatment status.
  - (b) In the design of regression discontinuity (RD), it's possible for nonlinearity to be confused with a discontinuity. To prevent this issue, what type of enhancement can you make to your model or what general suggestion would you offer?